

Angular Hands-On 10

Integration, Testing & Deployment

Objectives

- Integrate all features from Hands-On 1 to 9 into a single **Student Course Portal** project.
- Perform unit testing and end-to-end testing.
- Ensure routing, forms, services, and NgRx state management work together.
- Prepare the application for deployment.

Tasks

Task 1: Integrate Features

- Merge components: Home, Course List, Enrollment Form (Template-Driven & Reactive), Student Profile.
- Ensure services (CourseService, EnrollmentService, ApiService) are consistent.
- Connect NgRx store to API service for course management.
- Apply directives and pipes across components.

Task 2: Configure Routing

- Finalize routes for all pages:
 - / → Home
 - /courses → Course List
 - /enroll → Template-Driven Form
 - /enroll-reactive → Reactive Form
 - /profile → Student Profile (protected by AuthGuard)

Task 3: Testing

- **Unit Testing:**
 - Write tests for services (mock API calls).
 - Test reducers and selectors in NgRx.
- **Component Testing:**
 - Verify form validation messages.
 - Check directive and pipe outputs.
- **End-to-End Testing (E2E):**

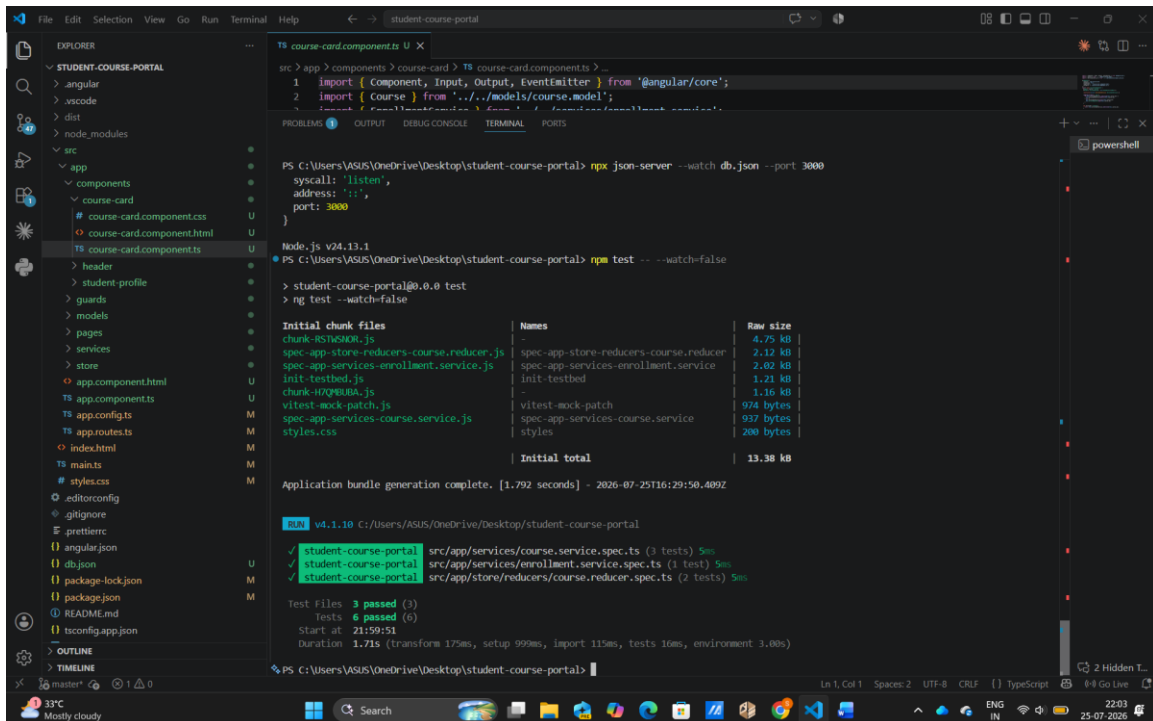
- Simulate user flow: login → enroll → view profile → logout.
- Use Protractor or Cypress for automation.

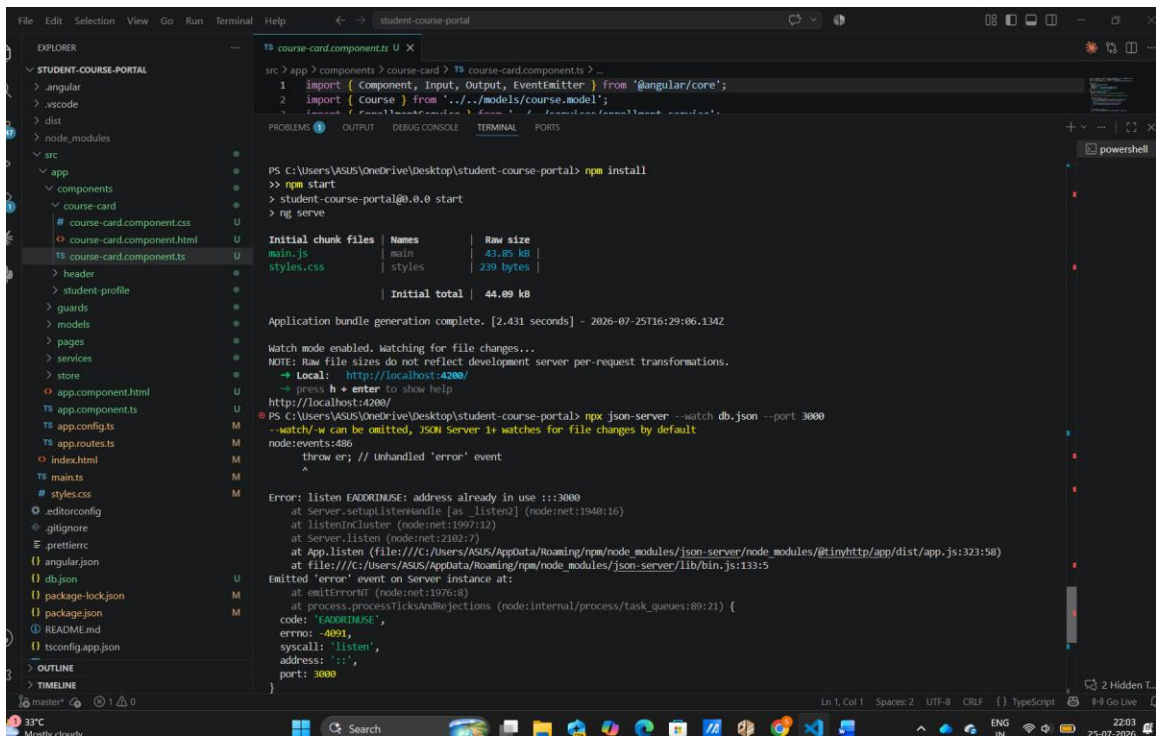
Task 4: Deployment Preparation

- Run production build:

```
ng build --prod
```

- Verify optimized bundle size.
- Deploy to a server (e.g., Firebase Hosting, Netlify, or local Apache/Nginx).





```
Node.js v24.13.1
PS C:\Users\ASUS\OneDrive\Desktop\student-course-portal> npm test -- --watch=false

> student-course-portal@0.0.0 test
> ng test --watch=false

Initial chunk files | Names | Raw size
chunk-RSTWSNOR.js | - | 4.75 kB |
spec-app-store-reducers-course.reducer.js | spec-app-store-reducers-course.reducer | 2.12 kB |
spec-app-services-enrollment.service.js | spec-app-services-enrollment.service | 2.02 kB |
init-testbed.js | init-testbed | 1.21 kB |
chunk-H7QMBUBA.js | - | 1.16 kB |
vitest-mock-patch.js | vitest-mock-patch | 974 bytes |
spec-app-services-course.service.js | spec-app-services-course.service | 937 bytes |
styles.css | styles | 200 bytes |
Initial total | 13.38 kB

Application bundle generation complete. [1.792 seconds] - 2026-07-25T16:29:50.409Z

RUN v4.1.10 C:/Users/ASUS/OneDrive/Desktop/student-course-portal

✓ student-course-portal src/app/services/course.service.spec.ts (3 tests) 5ms
✓ student-course-portal src/app/services/enrollment.service.spec.ts (1 test) 5ms
✓ student-course-portal src/app/store/reducers/course.reducer.spec.ts (2 tests) 5ms

Test Files 3 passed (3)
Tests 6 passed (6)
Start at 21:59:51
Duration 1.71s (transform 175ms, setup 999ms, import 115ms, tests 16ms, environment 3.00s)

PS C:\Users\ASUS\OneDrive\Desktop\student-course-portal>
```

